



**UTM**  
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**SECV2113 HUMAN COMPUTER INTERACTION**

**SECTION 01**

**ASSIGNMENT 1**

**INDUSTRY INNOVATION AND INFRASTRUCTURE**

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## Introduction

In recent years, there has been a noticeable trend in supermarkets where self-checkout payment options are increasingly being introduced at checkout counters. It is undeniable that this self-checkout system brings numerous benefits to consumers.



Firstly, self-checkout provides convenience by allowing consumers to scan the barcodes of the products and make payment independently, thereby reducing wait times, particularly for those with only a handful of items to purchase. This enhances the overall shopping experience of consumers, making their purchases more efficient.

Additionally, the implementation of self-checkout systems has led to a reduction in the number of staffed checkout lanes. While this may seem like a cost-cutting measure for supermarkets, it can also lead to potential savings for consumers, as operational efficiencies can sometimes result in lower prices or improved service. The funds saved from staffing reductions can improve various aspects of the shopping experience, whether it's through store renovations, expanding product offerings, or enhanced customer service efforts.

Furthermore, using the self-checkout method also allows consumers to go completely cashless, unlike some traditional counters that only accept cash payments. With self-service payment, consumers can use various payment methods such as credit cards and electronic wallets. Therefore, they no longer need to carry cash when going out, reducing the risk of theft.

Lastly, this system promotes contactless checkout options, which are becoming more popular among consumers and are prioritised for health and safety in today's environment. By reducing physical contact between consumers, supermarket staff, and other shoppers, self-service payment not only makes things more convenient but also emphasises safety measures, lowering the risk of spreading germs.

In essence, self-checkout represents a modernization of the retail experience, providing consumers and supermarkets with a more streamlined and efficient process than traditional over-the-counter payment methods.

## Features that Persuade User to Execute “Target Behaviour”

In a self-checkout system, the most important thing that is required from the user is to scan the barcode of the products they wish to purchase and make the payment. To induce users to execute the target behaviour, sufficient motivation, ability and trigger is required.

### Scan Barcode

The user must scan the barcode of the products they want to purchase using a barcode scanner. This allows the system to identify the product and its price, thus performing the calculations needed.

### Motivation

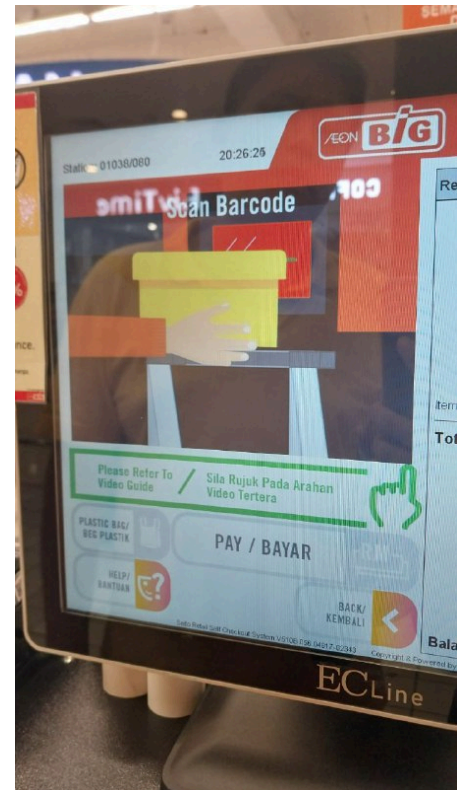
Users are motivated to buy the item and pay for it. To let the system know the user intent, the user must scan the barcode and add the product to the cart.

### Ability

The barcode scanner is provided at the self-checkout system and the barcode is printed on each item. The behaviour can be done by anyone that is tall enough to reach the scanner and can see the printed barcode. Furthermore, the process is intuitive and requires minimal instructions.

### Trigger

The text on the screen of the self-checkout system prompts the user to scan the barcode of the products. Additionally, the scanning head of the barcode scanner would shine LED or laser light to detect the barcode, indicating where to scan and triggering the action. Other than that, when an item is successfully scanned, a “beep” will sound that signals the user to take the next action, either to scan another item or to proceed to payment.



*Fig 1 Prompt on Self-checkout System*

## **Make Payment**

The user needs to select a payment method and complete the payment, either by QR payment or card payment.

### ***Motivation***

Users must pay for the items to acquire them. The user may experience pleasure from acquiring the items purchased, or in some cases, pain if the user is frugal.

### ***Ability***

The screen will first display the payment method that can be selected by the user. The users need to have cards issued by the service providers (e.g. Visa, MasterCard, UnionPay) or e-wallets (e.g. TouchNGo, Grab) that are accepted by the merchant. The merchant provides the card payment machine, and users must carefully follow instructions to complete the payment. This activity requires little thinking for first-time users as they are not familiar with the process.

### ***Trigger***

If the user selects to pay by card, the card payment machine will provide instructions on its screen and the user would need to respond accordingly. On the other hand, if the user selects the QR payment, the QR scanner will light up the LED or laser to detect the QR code, indicating where to scan and triggering the action.



*Fig 2 User scanning the QR code using a barcode scanner*

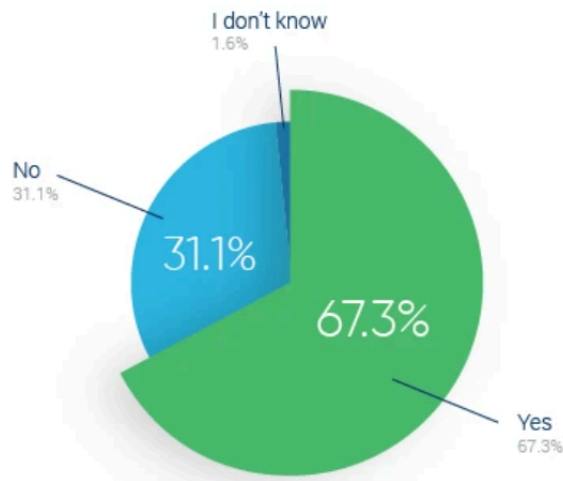
## Demonstration Video

[https://drive.google.com/file/d/1MjSBFCM7ekzfjHYi1iQqW6H\\_NuIl6ftx/view?usp=sharing](https://drive.google.com/file/d/1MjSBFCM7ekzfjHYi1iQqW6H_NuIl6ftx/view?usp=sharing)

### Failure to Persuade a Target Behaviour

However, the self-checkout technology uses a complicated procedure that may confound users, particularly the elderly. For example, in terms of user interface complexity, self-checkout systems frequently feature touchscreen displays, scanners, and multiple buttons, which can be hard for certain users, particularly those who are unfamiliar with technology. Users require time to learn how to navigate the various displays and options, and become familiar with the system. According to the pie chart below, 67.3% of users are unable to successfully use the self-checkout system. [1] Therefore, it could be better if users could just grab and go when making a purchase instead of having to scan the barcode.

? Have you ever tried to use a self-service checkout and had it not work for you?



In addition, there are usually a few steps in the checkout process: scanning the items, bagging them, choosing a payment method, and confirming the transaction. Users may need to follow a sequential set of instructions for each step. But mistakes will happen when you're checking out, such as improperly scanned items, problems with your credit card, or mishandled bags. Now, users might have to troubleshoot these difficulties on their own, which can be difficult if they don't know how to fix them. There was a scenario where the user dropped her card in the crack of the self-checkout devices. [2] This situation can be improved if there was technology that could recognize the things purchased by the user and automatically complete the payment when the user exits the store.



*Diagram shows that the staff is helping the customer to find her card.*

Furthermore, self-checkout machines may have technological faults or glitches, such as frozen screens or faulty scanners, complicating the checkout procedure. However, unlike traditional checkout lines, where a cashier is reachable to assist, self-checkout systems may not always have people on site to assist users who face difficulties or have problems.

In general, self-checkout systems are efficient and convenient, but they can also be difficult for users who are not advanced in technology or who run into unexpected issues when checking out. Improving the design, providing clear instructions, and offering assistance can help to reduce some of these difficulties and make the experience more user-friendly for everyone.

### Reference

- [1] “The State of Self-Service Checkouts,” *Raydiant*, Feb. 19, 2021.  
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- [2] N. Zucker, “CARD CHAOS,” *The US Sun*, Apr. 26, 2023. [Online]. Available:  
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